

Education

University of Florida

Bachelor of Science - Computer Science

January 2023 – May 2025

Gainesville, FL

- 3.98 GPA — Dean's List
- 2024 Barry Goldwater Scholar
- AI² Center Student Club Co-Director of Career Development and Future of Work Committee

The Honors College at Miami Dade College

Associate in Arts - Computer Science

August 2020 – July 2022

Kendall, FL

- 4.0 GPA — Dean's List
- Recipient of Honors College Fellows Award
- Vice President of the Computer Club

Research Experiences

Modern AI and Learning Technologies Lab

University of Florida

January 2023 – Present

Gainesville, FL

- Developed near-miss traffic incident identification software for the Broward County Mobility Advancement Program under grant number BCF #202 AGR00025372 made possible by the Penny for Transportation
- Trained a YOLOv8 (You Only Look Once) model with convolutional neural networks in Python to detect dangerous traffic patterns and classify vehicles with 99% accuracy in Dr. Sanjay Ranka's lab

National Renewable Energy Laboratory — SULI Intern

Department of Energy

August 2024 – December 2024

Golden, CO

- Compared two carbon footprint calculation methods for the Open Platform for Agile Trip Heuristics (OpenPATH) project: a baseline naive approach versus a sophisticated model incorporating National Transit Database data
- Coded a Python API for OpenTripPlanner to model human choice behavior in transportation mode selection

Computing for Global Challenges REU — Biocomplexity Institute

University of Virginia

May 2022 – January 2024

Charlottesville, VA

- Optimized a machine learning earthquake forecasting model to a Normalized Nash-Sutcliffe Efficiency of 98.3%
- Programmed a batch hyperparameter experiment generator on the Rivanna supercomputer to find the best meteorological cloud masking model

FGLSAMP Data Science and AI REU with nanoBIO Hub

Florida A&M University

June 2021 – December 2021

Tallahassee, FL

- Learned data science fundamentals in the Florida Georgia Louis Stokes Alliance for Minority Participation program
- Predicted blockchain-based cryptocurrency price changes within \$0.229 USD accuracy using a long short-term memory model in keras

Publications

- [1] **Jacques Fleischer**, Heriberto Lopez, Kenneth Krause, Tania Banerjee, Min-Tang Li, Anand Rangarajan, Sanjay Ranka. "That Was Close: A Video-Based Safety Assessment of Signalized Traffic Intersections along a High-Speed Corridor". In: *Proceedings of the Transportation Research Board 2025 Annual Meeting*. Accepted for presentation. 2025.
- [2] Abigail R. Wheelis, Natalie Schultz, Jack Greenlee, **Jacques Fleischer**, Louis Grassi, K. Shankari. "CHEER: Calculating Holistic Energetic Emissions on Route — A Fine-Grained, Quasi-Real-Time System for Computing the Energy and Emission Impact of Personal Transportation". Submitted to the National Bureau of Economic Research Conference on Data Infrastructure for Decarbonizing Transportation. 2025.
- [3] **Jacques Fleischer**, Ryan Pallack, Ahan Mishra, Gustavo Riente de Andrade, Subhadipto Poddar, Emmanuel Posadas, Robert Schenck, Tania Banerjee, Anand Rangarajan, Sanjay Ranka. "Video-based Pedestrian and Vehicle Traffic Analysis During Football Games". In: *Proceedings of the 27th IEEE International Conference on Intelligent Transportation Systems (ITSC)*. 2024.
- [4] Gregor von Laszewski, **Jacques Fleischer**, Geoffrey C. Fox, Juri Papay, Sam Jackson, Jeyan Thiyagalingam. "Templated hybrid reusable computational analytics workflow management with Cloudmesh, applied to the deep learning MLCommons Cloudmask application". In: *2023 IEEE 19th International Conference on e-Science (e-Science)*. IEEE. 2023, pp. 1–6.

- [5] Gregor von Laszewski, **Jacques Fleischer**, Robert Knuuti, Geoffrey C. Fox, Jake Kolessar, Thomas S. Butler, Judy Fox. “Opportunities for enhancing MLCommons efforts while leveraging insights from educational MLCommons earthquake benchmarks efforts”. In: *Frontiers in High Performance Computing* 1 (2023).
- [6] **Jacques Fleischer**, Gregor von Laszewski, Carlos Theran, Yohn Jairo Parra Bautista. “Time series analysis of blockchain-based cryptocurrency price changes”. In: *2022 Emerging Researchers National (ERN) Conference in STEM*. The conference was postponed/canceled due to the COVID-19 pandemic and emerging variants. Washington, D.C., 2022.
- [7] **Jacques Fleischer**, Gregor von Laszewski, Carlos Theran, Yohn Jairo Parra Bautista. “Time series analysis of cryptocurrency prices using long short-term memory”. In: *Algorithms* 15.7 (2022), p. 230.
- [8] Gregor von Laszewski, **Jacques Fleischer**, Geoffrey C. Fox. “Hybrid reusable computational analytics workflow management with Cloudmesh”. arXiv preprint arXiv:2210.16941. 2022.